Merge Two Binary Trees (medium)

Problem Statement

Given two binary trees, `root1` and `root2`, merge them into a single, new binary tree.

If two nodes from the given trees share the same position, their values should be summed up in the resulting tree. If a node exists in one tree but not in the other, the resulting tree should have a node at the same position with the value from the existing node.

Examples

Example 1:

Trees:

```
Tree 1:      1             Tree 2:      1
           /   \                      /   \
          3     2                    2     3

Merged:       2
            /   \
           5     5
```

Justification: The root nodes of both trees have the value `1`, so the merged tree's root has a value of `1 + 1 = 2`. For the left child, `3 + 2 = 5` and for the right child, `2 + 3 = 5`.

Example 2:

Trees:

```
Tree 1:      5             Tree 2:      3
           /   \                      /   \
          4     7                    2     6

Merged:       8
            /   \
           6    13
```

Justification: The root nodes have values `5` and `3` respectively. So, the merged tree's root value becomes `5 + 3 = 8`. The left child is `4 + 2 = 6` and the right child is `7 + 6 = 13`.

Example 3:

Trees:

```
Tree 1:      2             Tree 2:      2
              \                       /
               5                     3

Merged:       4
            /   \
           3     5
```

Justification: The root nodes have values `2` each, so the merged tree's root value is `2 + 2 = 4`. Tree 1 doesn't have a left child, so we take the left child of Tree 2 as it is, which is `3`. Similarly, Tree 2 doesn't have a right child, so the merged tree's right child is the same as Tree 1, which is `5`.

Example 4:

Trees:

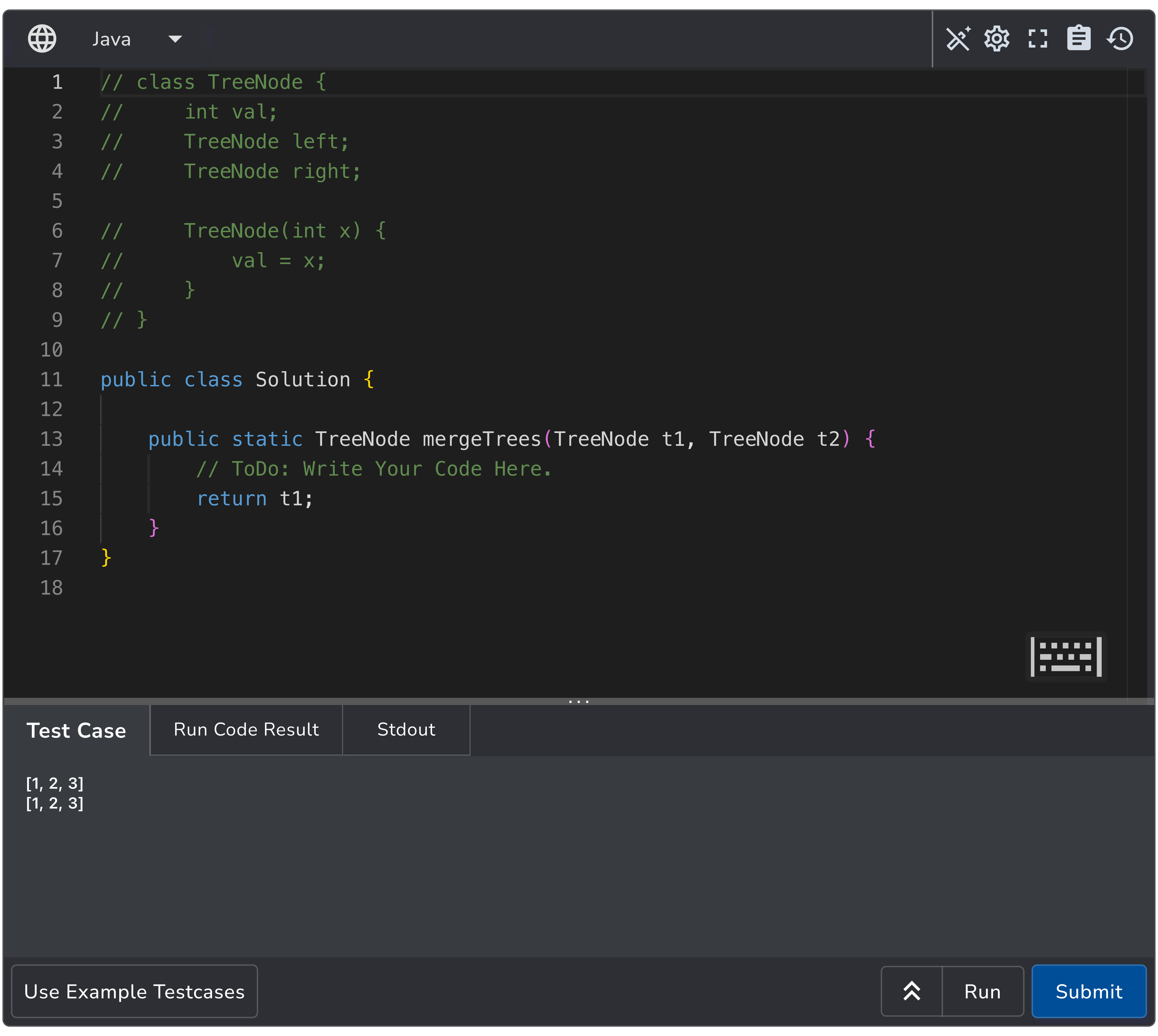
```
Tree 1:      10            Tree 2:      10
           /    \                     /    \
          5     15                   6     16

Merged:       20
            /    \
           11    31
```

Justification: The root nodes have the value `10`, so they add up to `20` for the merged tree. The left child values add up to `5 + 6 = 11` and the right child values sum up to `15 + 16 = 31`.

Try it yourself

Try solving this question here:



```java
// class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;

//     TreeNode(int x) {
//         val = x;
//     }
// }

public class Solution {

    public static TreeNode mergeTrees(TreeNode t1, TreeNode t2) {
        // ToDo: Write Your Code Here.
        return t1;
    }
}

```

```
[1, 2, 3]
[1, 2, 3]
```

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